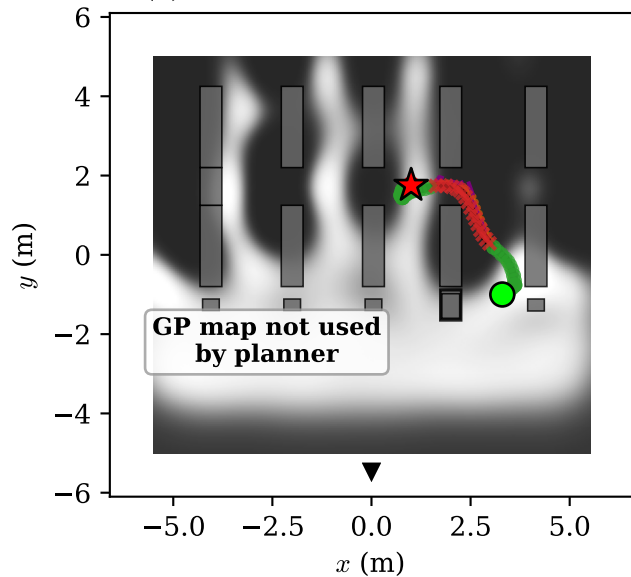


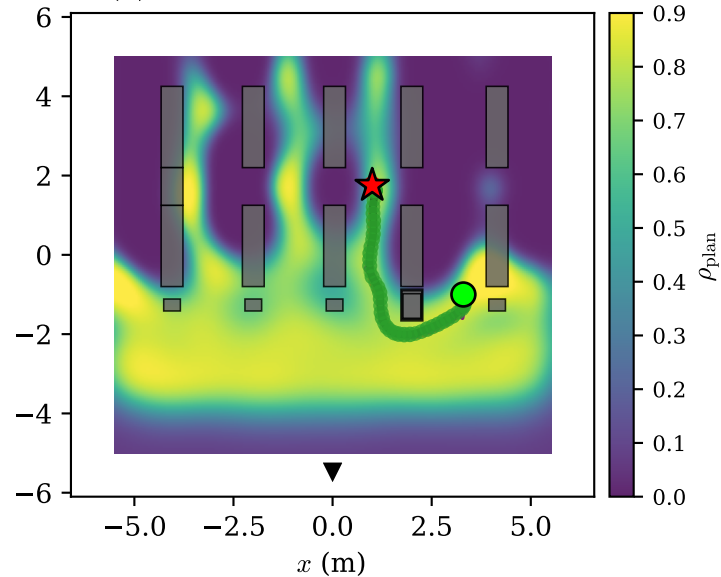
# Paired representative run on the route-choice task (seed 0)

● start ★ goal ▼ camera — executed truth - - belief mean ● YOLO detection × YOLO miss

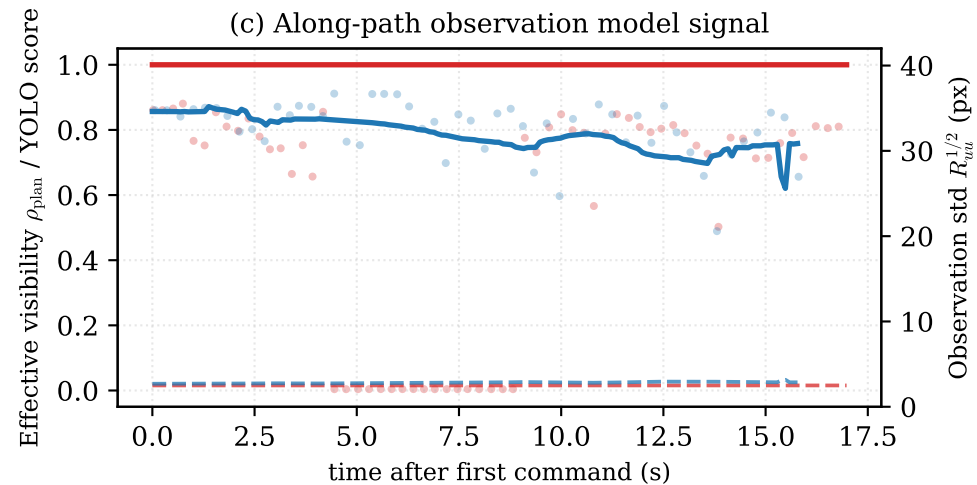
(a) Constant covariance — seed 0



(b) Learned covariance — seed 0

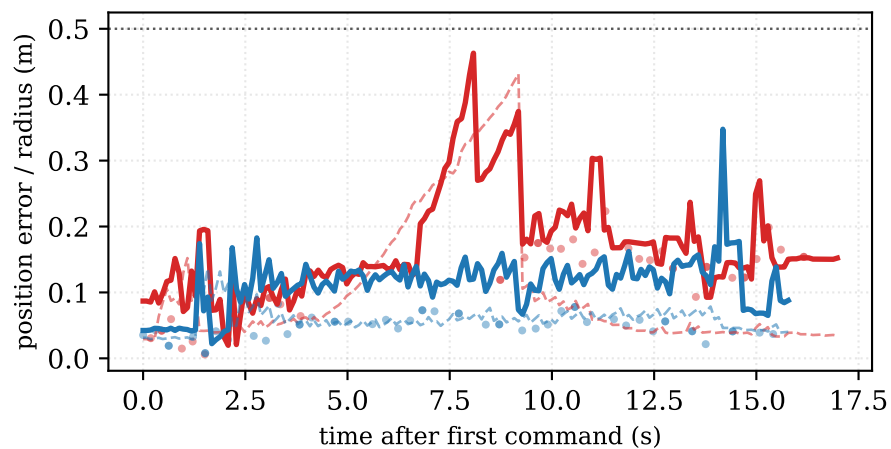


(c) Along-path observation model signal



— C1  $\rho_{\text{plan}}$  ● C1 YOLO - - C1  $R_{uu}^{1/2}$   
— C2  $\rho_{\text{plan}}$  ● C2 YOLO - - C2  $R_{uu}^{1/2}$

(d) Truth-belief error with  $2\sigma$  belief radius



— C1  $e_p$  (mean 0.17 m) - - C2  $r_{2\sigma}$   
- - C1  $r_{2\sigma}$  ● C2 camera update  
● C1 camera update - - - 0.5 m ref  
— C2  $e_p$  (mean 0.11 m)